The own count with its conditional variance allocation retained

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The difficult negative phases should not be bounded independently of the incoming lower tail. Positive conditional noise is unfavorable for the own count, whereas negative conditional noise is unfavorable for that incoming tail. The elementary bound below retains this distinction. It is a local estimate, not a completed count comparison above noise $1/50$.

Let U_0, U_1 be centered laws with

$$-A \le U_0 \le a_0, \qquad -a_1 \le U_1 \le A, \qquad A > 0, \quad 0 \le a_0, a_1 \le A.$$

Write $v_j = \mathbb{E}U_j^2$, $\mu_j = \mathbb{E}U_j^3$. For an integer $n \ge 1$ and $0 \le k \le n$, fix the legal counts $N_0 = n - k$ and $N_1 = k$, and let W be the sum of N_j independent variables of each type; all variables are independent. Set $T_j = N_j v_j$ and $t = T_0 + T_1$. When marginal quantities are used, take $0 < p < 1$, $q = 1 - p$, $B \sim \text{Bernoulli}(p)$, and give U the conditional law U_j given $B = j$. Then $s = \mathbb{E}U^2 = qv_0 + pv_1$ and $\lambda = ns$.

Lemma 1 (A conditional fourth/third-moment bound)**.** *For every real κ,*

$$\mathbb{E}W^4 - \kappa\mathbb{E}W^3 \le c_0(\kappa)T_0 + c_1(\kappa)T_1 + 3t^2,$$

where

$$c_0(\kappa) = \max\{A^2 + \kappa A, a_0^2 - \kappa a_0\}, \qquad c_1(\kappa) = \max\{A^2 - \kappa A, a_1^2 + \kappa a_1\}.$$

The right side can be reduced by $3(N_0v_0^2 + N_1v_1^2)$ if that correction is useful separately.

Proof. The convex function $x^2 - \kappa x$ attains its maximum on a closed interval at an endpoint. Multiplication by $x^2 \ge 0$ and integration give $\mathbb{E}U_j^4 - \kappa\mathbb{E}U_j^3 \le c_j(\kappa)v_j$. Centering and independence give

$$\mathbb{E}W^3 = N_0\mu_0 + N_1\mu_1, \qquad \mathbb{E}W^4 = \sum \mathbb{E}U_i^4 + 3t^2 - 3\sum(\mathbb{E}U_i^2)^2.$$

Substitution proves both statements. □

This improves the separate bounds $\mathbb{E}W^4 \le A^2t + 3t^2$ and $|\mathbb{E}W^3| \le At$: each $c_j(\kappa)$ is at most $A^2 + A|\kappa|$. For nondegenerate support intervals the conditional constants are asymptotically sharp. For example, a centered two-point law on $\{-\varepsilon, A\}$ has

$$\frac{\mathbb{E}Z^4 - \kappa\mathbb{E}Z^3}{\mathbb{E}Z^2} = A^2 - A\varepsilon + \varepsilon^2 - \kappa(A - \varepsilon) \longrightarrow A^2 - \kappa A.$$

The other three endpoints are obtained by reflection or by putting the rare atom at an inner endpoint. This sharpness concerns the linear per-summand moment bound, not an extremal convolution theorem.

Proposition 2 (An own-count bound for any quartic minorant)**.** *Suppose*

$$h(x) = \frac{-x^4 + b_3x^3 + b_2x^2 + b_1x + b_0}{M} \le \mathbf{1}_{(0,\infty)}(x), \qquad M > 0.$$

For $r \ge 0$ define

$$\kappa = b_3 - 4r, \quad L = b_2 + 3b_3r - 6r^2, \quad K = Mh(r).$$

Then

$$\mathbb{P}(W > -r) \ge \frac{Lt - 3t^2 - c_0(\kappa)T_0 - c_1(\kappa)T_1 + K}{M}. \tag{1}$$

The bound includes atoms at the threshold: the event is strict.

Proof. Since $\mathbb{E}W = 0$, expansion gives $M\mathbb{E}h(W + r) = -\mathbb{E}W^4 + \kappa\mathbb{E}W^3 + Lt + K$. Apply Lemma 1 and the pointwise minorant. □

For the earlier quartic

$$h_\alpha(x) = \frac{-x^4 + 3\alpha^2x^2 + 2\alpha^3x}{5\alpha^4},$$

the numerator factors as $-x(x + \alpha)^2(x - 2\alpha)$; it is nonpositive for $x \le 0$ and its positive maximum is $(9 + 6\sqrt{3})\alpha^4/4 < 5\alpha^4$. Hence it is a valid minorant. Here $\kappa = -4r$. If $a_0 + a_1 \le d \le A$ and $0 \le r \le (A - d)/4$, then

$$c_0 = A^2 - 4Ar, \qquad c_1 = A^2 + 4Ar.$$

Indeed the first comparison is equivalent to $4r \le A - a_0$; the second follows from $A^2 + 4Ar \ge a_1^2 - 4ra_1$. Consequently

$$\begin{aligned}\mathbb{P}(W > -r) \ge \frac{1}{5\alpha^4}\{&(3\alpha^2 - A^2 - 6r^2)t - 3t^2 + 4Ar(T_0 - T_1) \\ &- r^4 + 3\alpha^2r^2 + 2\alpha^3r\}.\end{aligned} \tag{2}$$

Compared with the separate fourth/third-moment estimate, the gain is exactly $8ArT_0/(5\alpha^4)$.

A material change at an admissible point. Take $A = 3/5$, $\alpha = 1/2$, $r = t = 3/40$, and $T_1 = 0$. After adding the normal-increment lower bound $-(21/20)r$, the new lower bound is $55479/800000 = .06934875$. The separate-moment bound is $-13641/800000 = -.01705125$. The gain is .0864. These parameters are realized by an actual law: take $n = 42$, $p = 1/2$, $k = 21$, $U_1 = 0$, and the centered law $U_0 \in \{-3/5, 1/168\}$. Then $ns = 3/40$, $T_0 = 3/40$, $a_0 + a_1 = 1/168 < .072$ and the corresponding raw threshold lies in $[-1.84, 1.84]$.

Normal slope through phase .1**.** Put $v = pq$, $\ell = (v + s)^{-1/2}$, $\pi_k = \binom{n}{k}p^kq^{n-k}$, and choose the phase by $np + H/\ell = k + u$. With $\delta = k - np$, the normal increment is

$$f(u) = \frac{\Phi((\delta + u)/\sqrt{n(v + s)}) - \Phi(\delta/\sqrt{n(v + s)})}{\pi_k}.$$

On $p \in [.4, .53]$, $n \ge 42$, $ns \le .075$ and $|H| \le 1.84$, one has $\sqrt{pq + s} < .502$. For $|u| \le .1$ this gives $|\delta| < 1.84 \cdot .502 + .1 = 1.02368 < 21/20$. The finite Stirling/logarithmic derivative estimate established in the v10 comparison, with $D = 21/20$ and $b = 1/10$, gives

$$n|\log f'(z)| \le \tfrac{8}{3}\lambda + \frac{2Db + b^2}{2(6/25)} + \frac{5D}{8} + \frac{3}{8} + \frac{D^3}{42}.$$

The exact right side at $\lambda = 3/40$, divided by 42, is less than $1/21$. Thus $.95 < f'(z) < 1.05$ on the whole integration segment. In particular $f(-r) \ge -(21/20)r$ for $0 \le r \le .1$. This invocation retains the finite Stirling proof and the fixed actual count; it is not a fixed-law asymptotic assertion.

Combining the own count and incoming tails

For fixed $n, p, k, \lambda > 0$, put $q = 1 - p$ and parametrize the allocation by $\eta = pv_1/s \in [0, 1]$, where $s = \lambda/n$. Then

$$v_0 = s(1-\eta)/q, \qquad v_1 = s\eta/p, \quad T_0 = N_0 s(1-\eta)/q, \quad T_1 = N_1 s\eta/p.$$

Fix the support caps and the quartic parameters independently of η. The right side of (1) is a concave function of η: all terms are affine except $-3t^2$. Endpoint tests may use the larger interval $[0, 1]$ even if some endpoint allocations are infeasible for the particular caps. They are then tests of a relaxation, not assumed laws.

For completeness, the needed incoming estimate can be proved without an additional probability theorem. For $h > 0$, $b > 0$, define

$$B_h(b) = \frac{e^{hb} - 1 - hb}{b^2} = h^2 \int_0^1 (1-z) e^{hzb}\, dz.$$

If Z is centered and $Z \le b$, the same integral remainder gives $\mathbb{E}e^{hZ} \le 1 + B_h(b)\mathbb{E}Z^2 \le e^{B_h(b)\mathbb{E}Z^2}$. Use $b = A$ for $-U_0$ and any fixed positive upper cap $b_1 \ge a_1$ for $-U_1$. For each legal incoming count $i \ge 1$,

$$\mathbb{P}(W_{k+i} \le u - i) \le \exp\{h(u-i) + B_h(A)(N_0 - i)v_0 + B_h(b_1)(N_1 + i)v_1\}. \tag{3}$$

Every exponent is affine in η, so their sum is convex. An infinite geometric extension is allowed if

$$h > (s/p) B_h(b_1).$$

Indeed the ratio is $\exp\{-h - B_h(A)v_0 + B_h(b_1)v_1\} < 1$ uniformly on $[0, 1]$. The added terms are positive even when their count is no longer legal. The infinite sum remains convex, being a pointwise limit of its convex partial sums.

It follows that an own-count lower bound minus a fixed nonnegative multiple of this incoming bound is concave in η. Its minimum is at an allocation endpoint. This is an exact reduction to two scalar estimates for each set of the other parameters. It does not itself prove either endpoint estimate.

The tilt and quartic may depend on the other parameters, but must be fixed while this allocation argument is applied. Taking a pointwise minimum over several incoming bounds need not preserve convexity. Likewise, retaining the favorable $3\sum v_i^2$ correction from Lemma 1 can destroy concavity in the allocation; dropping it is deliberate here.

Retaining the cubic repair in the same inequality

Suppose additionally $a_0 < p$, $a_1 < q$, and set

$$\ell = (pq + s)^{-1/2}, \quad \rho = \mathbb{E}|\ell(B + U - p)|^3, \quad \rho_B = (1 - 2pq)/\sqrt{pq}.$$

The signs are fixed, giving the exact identity

$$\rho = \ell^3\{pq(1-2pq) + 3pq(v_0 + v_1) + p\mu_1 - q\mu_0\}.$$

Let $m > 0$, $C \ge 0$, and assume $N_0, N_1 > 0$. Define

$$\gamma = C\ell^3 M/m, \quad \kappa_0 = \kappa - \gamma q/N_0, \quad \kappa_1 = \kappa + \gamma p/N_1.$$

The same proof, now combining the third moments before bounding them, gives

$$m\mathbb{P}(W > -r) + C(\rho - \rho_B) \geq \frac{m}{M}\{Lt - 3t^2 - c_0(\kappa_0)T_0 - c_1(\kappa_1)T_1 + K\} + C\big[\ell^3\{pq(1-2pq) + 3pq(v_0+v_1)\} - \rho_B\big].$$

For fixed n, p, k, λ, m, C this is still concave in η. In the comparison, $m = \sqrt{n}\,\pi_k$ is independent of η. Outward replacements for m must respect the sign of the complete expression; one cannot replace it separately inside κ_j without checking that direction. This bound retains the same third moments in both the own-count and marginal-moment terms.

An optional sharper bound at fixed summand variance

The linear coefficient bound has a useful refinement, though it is not needed for the endpoint reduction. If $Z \in [-b, A]$ is centered with variance v and

$$A - b - 2v/A - \kappa \geq 0,$$

then

$$\mathbb{E}(Z^4 - \kappa Z^3) \leq v\{A^2 - v + v^2/A^2 - \kappa(A - v/A)\}.$$

To prove this, subtract $(x - A)(x + v/A)^2(x + A - 2v/A - \kappa)$ from $x^4 - \kappa x^3$. The resulting polynomial is quadratic and majorizes the original on $[-b, A]$. Its expectation depends only on the mean and variance; evaluate it on the centered law supported at $-v/A, A$. This law has the required variance, and $v/A \leq b$ follows from $v \leq Ab$. At both its atoms the subtraction vanishes. The case $v = 0$ is immediate. Reflection gives the analogous negative-endpoint estimate. This refinement should not be inserted into the allocation argument without rechecking its curvature.

Status. The lemmas are complete local deductions. Exact rational checks cover the normal-slope constants, moment identities, support endpoints, actual conditional laws and illustrative improvement. No complete comparison on $.02 < \lambda \leq .075$ is claimed. The remaining work is to prove the two allocation-endpoint comparisons with one common phase-dependent choice of tilt and own-count bound.